

Stern-Gerlach Experiment

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27/11/2025

Stern-Gerlach Experiment

Historical Context

Planck quanti

Hello world

einstein photoe

Hello world

bohr atom

Hello world

Experimental Apparatus

Angular & Magnetic momenta

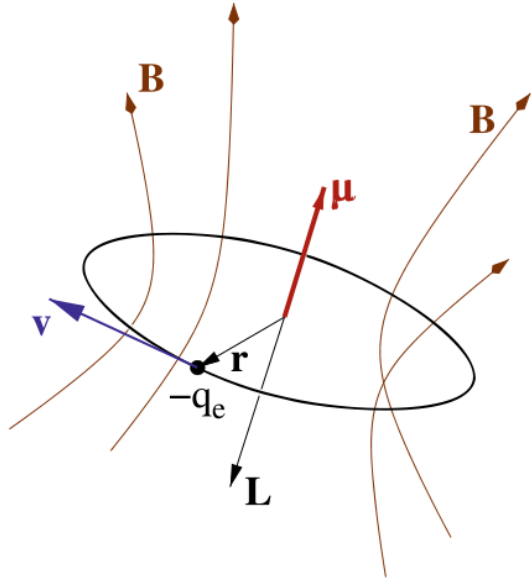
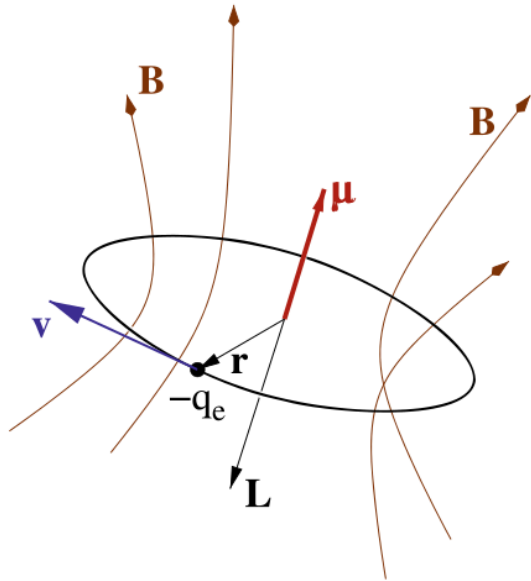


Figure 1: **Angular** and **magnetic** momenta generated by an electron orbiting circularly.

Angular & Magnetic momenta

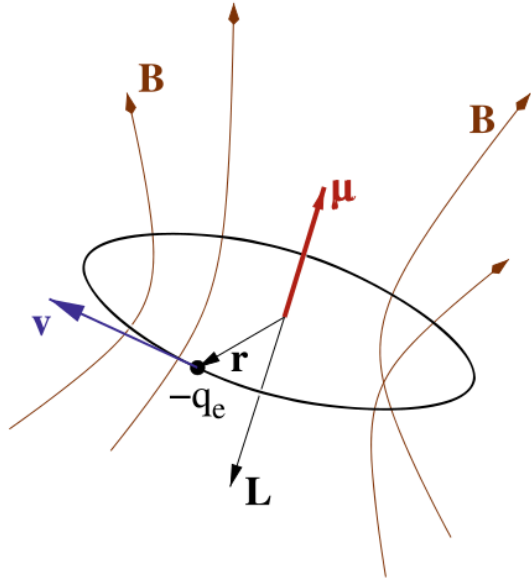


Angular momentum $\mathbf{L} = \mathbf{r} \times \mathbf{p}$

Magnetic momentum $\boldsymbol{\mu} = I \Sigma \hat{\mathbf{n}}$

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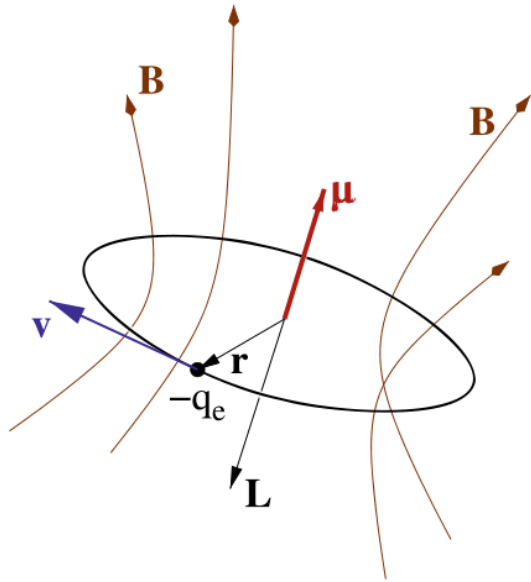
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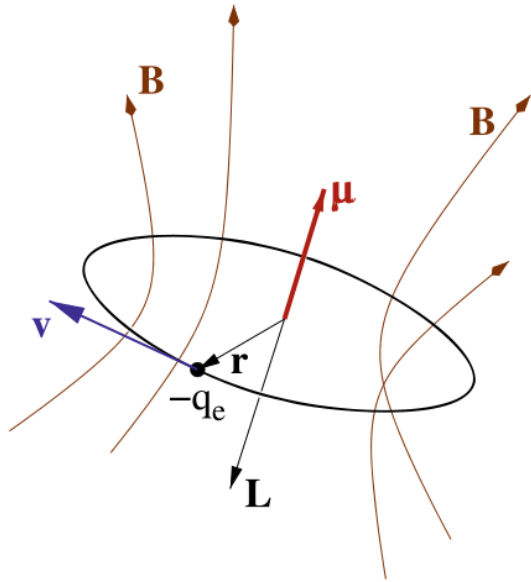


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The **interaction** between $\boldsymbol{\mu}$ and an **external magnetic field** $\mathbf{B}$ is represented by the magnetic hamitonian

$$H_{\text{magn}} = -\boldsymbol{\mu} \cdot \mathbf{B}$$

Stern-Gerlach Experiment Apparatus

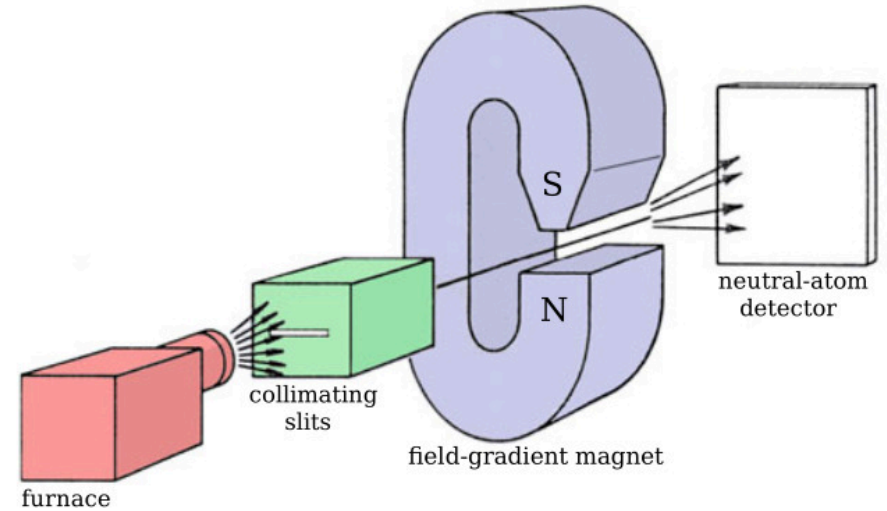
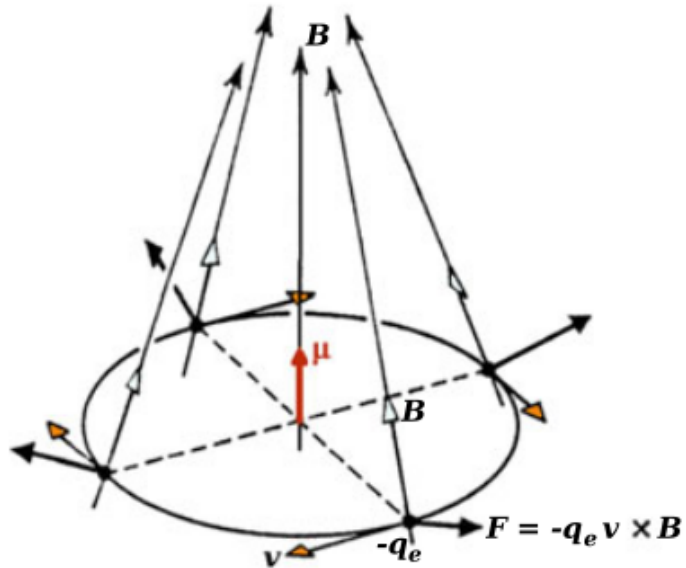


Figure 2: Stern-Gerlach apparatus.

Stern-Gerlach Experiment

Magnetic Interaction



$$F_{\text{magn}} = -\nabla H_{\text{magn}} = \nabla(\mu \cdot B)$$

Figure 3: Microscopic origin of the magnetic force.

Stern-Gerlach Experiment

Magnetic Interaction

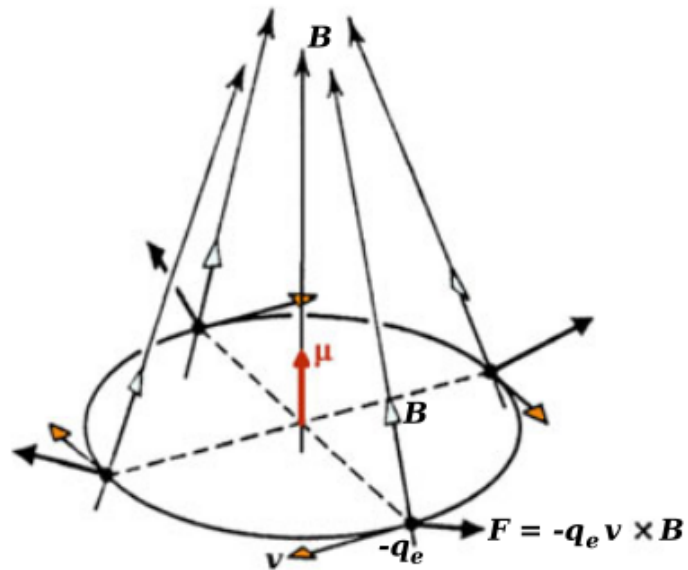


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Stern-Gerlach Experiment

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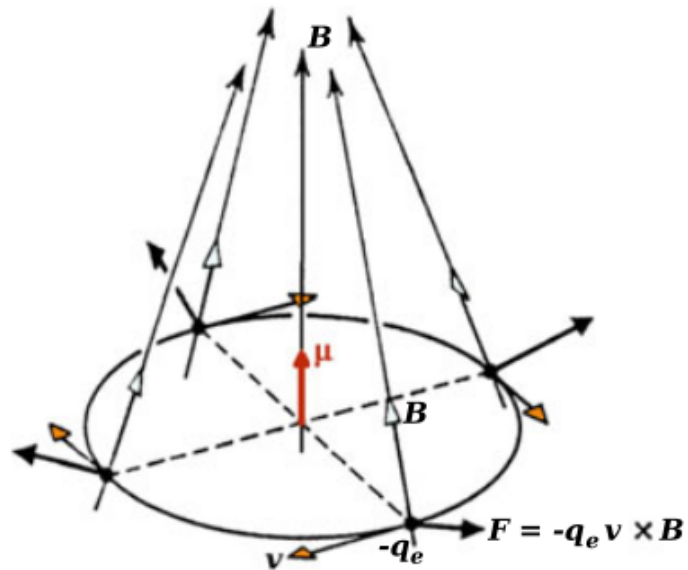


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Stern-Gerlach Experiment

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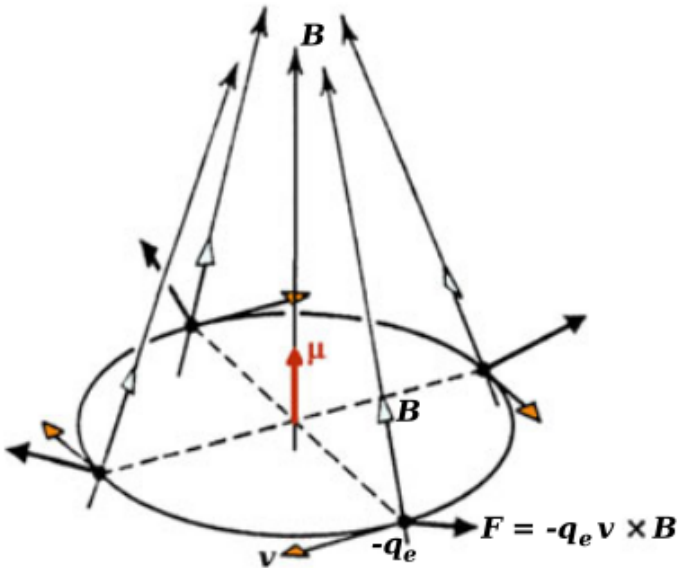


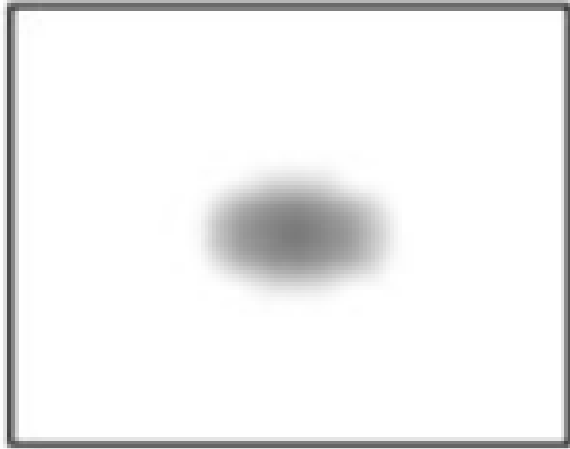
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$$F_z = \mu_z \frac{\partial B_z}{\partial z} = m\ddot{z}$$

$$z(t) = z_0 + \dot{z}_0 t + \frac{\mu_z}{2m} \frac{\partial B_z}{\partial z} t^2$$

Experimental Outcome

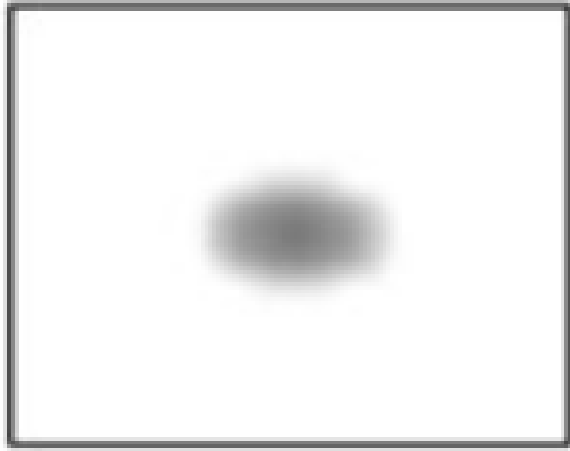


classically predicted



observed

Experimental Outcome



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Why do we see two deflected lumps?

Quantum Mechanics

QM — States

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$$|\psi\rangle = \sum_i |e_i\rangle \langle e_i | \psi \rangle = \sum_i c_i^\psi |e_i\rangle$$

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$$\langle \varphi | \psi \rangle = \sum_i \langle \varphi | e_i \rangle \langle e_i | \psi \rangle = \sum_i (c_i^\varphi)^* c_i^\psi$$

- **Born's rule**

$$\langle \psi | \psi \rangle = \sum_i (c_i^\psi)^* c_i^\psi = \sum_i |c_i^\psi|^2 = 1$$

$|c_i^\psi|^2$ is the **probability** to find $|\psi\rangle$ in $|e_i\rangle$

QM — Operators

A **linear operator** $\hat{O}$ is a transformation from an Hilbert space into itself, that acts like

$$\hat{O}(\alpha|\psi\rangle + \beta|\varphi\rangle) = \alpha\hat{O}|\psi\rangle + \beta\hat{O}|\varphi\rangle = \alpha|\psi'\rangle + \beta|\varphi'\rangle$$

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The action of $\hat{O}$ on a state can be entirely represented by its action on some **basis**:

$$|\psi'\rangle = \hat{O}|\psi\rangle = \sum_i \hat{O}|e_i\rangle\langle e_i|\psi\rangle = \sum_{ij} |e_j\rangle\langle e_j|\hat{O}|e_i\rangle\langle e_i|\psi\rangle = \sum_{ij} |e_j\rangle O_{ij} c_i^\psi$$

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We can also define the **adjoint** operator $\hat{O}^\dagger$: $\langle\psi'| = \langle\psi|\hat{O}^\dagger$ $\hat{O}_{ij}^\dagger = (\hat{O}_{ji})^*$

QM — Operators & Observables

We can associate every **observable** of a system to an operator

$$\hat{O} = \sum_i \lambda_i |e_i\rangle \langle e_i| \quad \hat{O}|e_i\rangle = \lambda_i |e_i\rangle$$

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where λ_i are the **possible results** and $|e_i\rangle$ are the **possible states** in which the system is founded **after** the measurement.

The requirements are that, **independently by the choice of the basis**,

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Such operators are called **hermitian** and satisfy $\hat{O}_{ij}^\dagger = \lambda_i^* \delta_{ij} = \lambda_i \delta_{ij} = \hat{O}_{ij}$

QM — Commutator of Operators

We define the **commutator** between two operators $\hat{A}$ and $\hat{B}$ as

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A} = -[\hat{B}, \hat{A}]$$

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Two observables can be **simultaneously** determined iff their operators are compatible, that is if they share a **common basis** of eigenstates:

$$\hat{A}|e_i\rangle = \lambda_i|e_i\rangle \quad \hat{B}|e_i\rangle = \mu_i|e_i\rangle \quad \Leftrightarrow \quad \langle e_i | [\hat{A}, \hat{B}] | e_j \rangle = (\lambda_i \mu_i - \lambda_j \mu_j) \langle e_i | e_j \rangle = 0$$

Position and impulse operators
don't commute!

$$[\hat{p}, \hat{x}] = -i\hbar$$

QM — Angular Momentum

Definition

$$\hat{L}_k = (\hat{\mathbf{x}} \times \hat{\mathbf{p}})_k = \varepsilon_{kij} \hat{x}_i \hat{p}_j$$

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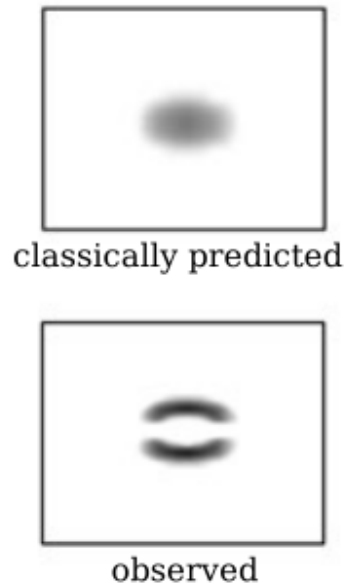
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Degeneracy $d_l = 2l + 1$

Stern-Gerlach Experiment

Experimental Outcome

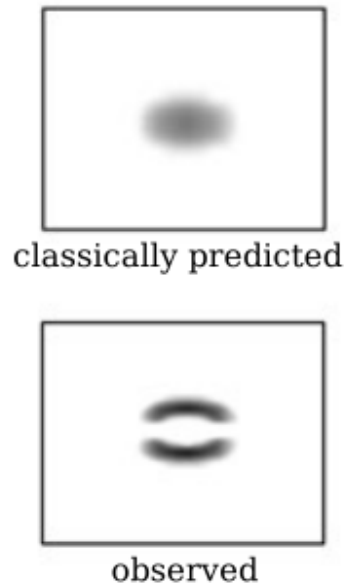


For an electron, $\hat{\mu} = \frac{\hbar q_e}{2m_e} \frac{\hat{L}}{\hbar}$

Figure 15: Stern-Gerlach theoretical versus experimental results.

Stern-Gerlach Experiment

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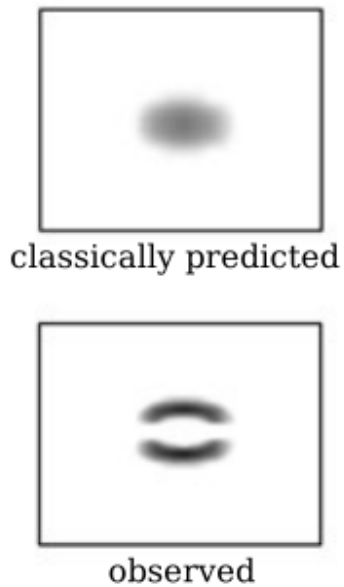


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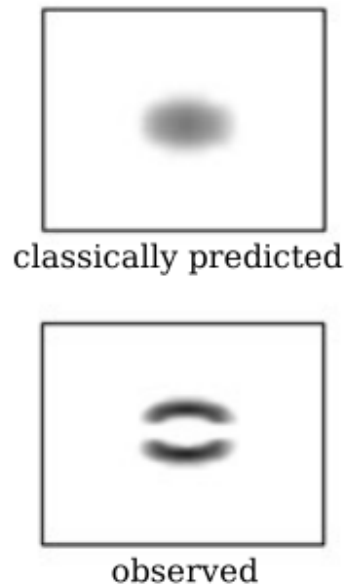


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The energy of the system $|\psi\rangle = |lm_l\rangle$ is

$$\langle \hat{H} \rangle_\psi = -\langle \hat{\mu}_z B_z \rangle_\psi = g_l \mu_B B_z \frac{1}{\hbar} \langle \hat{L}_z \rangle_\psi$$

$\hat{L}_z$ exhibits an odd number of discrete eigenvalues $(2l + 1)$, so...

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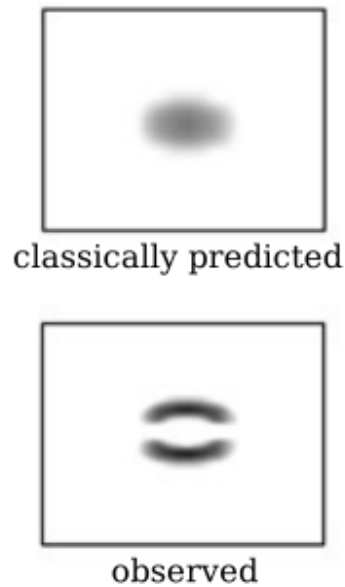


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Why do we see **two deflected lumps?**

Spin

QM — Intrinsic Angular Momentum

Commutation rules

$$[\hat{S}_i, \hat{S}_j] = i\hbar\epsilon_{ijk}\hat{S}_k \quad [\hat{S}_i, \hat{S}^2] = 0$$

Basis

The common eigenstates of $\hat{S}^2$ and $\hat{S}_z$ are $|sm_s\rangle$.

Spectrum

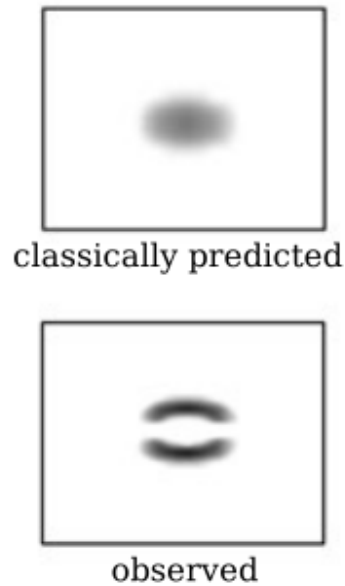
$$\hat{S}^2|sm_s\rangle = \hbar^2 s(s+1)|sm_s\rangle$$

$$\hat{S}_z|sm_s\rangle = \hbar m_s|sm_s\rangle \quad -s \leq m_s \leq s$$

Degeneracy $d_s = 2s + 1$

Stern-Gerlach Experiment

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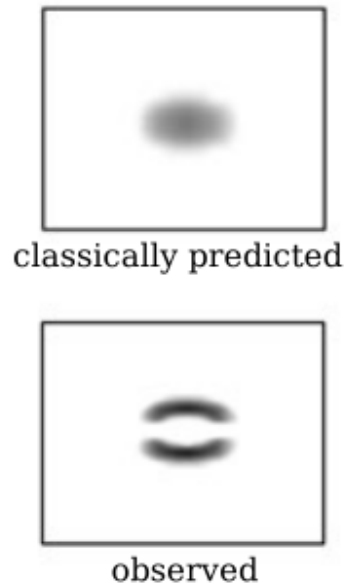
Total Magnetic Momentum

$$\hat{\mu} = \hat{\mu}_l + \hat{\mu}_s = -\frac{\mu_B}{\hbar} (g_l \hat{L} + g_s \hat{S})$$

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Stern-Gerlach Experiment

Experimental Outcome



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Counting the spots on the screen:

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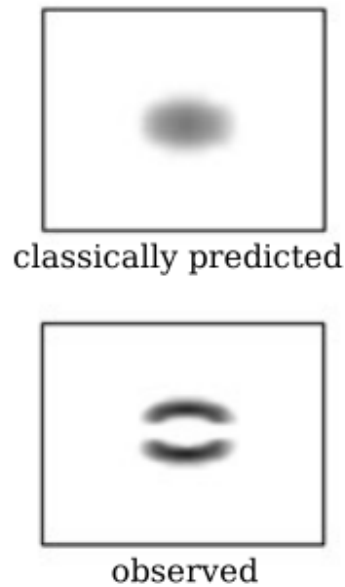


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Also, to make the deflection magnitude match the experiment $g_s = 2$

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$$\hat{\mathbf{J}}^2|jm_j\rangle = \hbar^2 j(j+1)|jm_j\rangle \quad |l-s| \leq j \leq l+s$$

$$\hat{J}_z|jm_j\rangle = \hbar m_j|jm_j\rangle \quad -j \leq m_j \leq j$$

Degeneracy

$$d_j = 2j + 1$$

Total Angular Momentum

Consider a state $|\psi\rangle = |lm_l\rangle \otimes |sm_s\rangle$

Definition

$$\hat{\mathbf{J}} = \hat{\mathbf{L}} + \hat{\mathbf{S}}$$

$$\hat{\mathbf{J}}^2 = \hat{\mathbf{L}}^2 + \hat{\mathbf{S}}^2 + 2\hat{\mathbf{S}} \cdot \hat{\mathbf{L}}$$

Commutation rules

$$[\hat{J}_i, \hat{J}_j] = i\hbar\epsilon_{ijk}\hat{J}_k \quad [\hat{J}_i, \hat{\mathbf{J}}^2] = 0$$

Basis

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Degeneracy

$$d_j = 2j+1 = (2l+1)(2s+1)$$

Coupling of Angular Momenta

Consider the magnetic field experienced by an e^- orbiting around a nucleus with Z p^+

$$\mathbf{B}(\mathbf{r}) = \frac{Zq_e}{4\pi\epsilon_0 c^2 m_e} \frac{\hat{\mathbf{L}}}{|\mathbf{r}|^3}$$

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Another option is the **coupled** basis

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Energy Spectra in Magnetic Field

$$\hat{H} = \hat{H}_0 + \hat{H}_{\text{s-o}} + \frac{\mu_B B_z}{\hbar} (\hat{L}_z + 2\hat{S}_z)$$

Energy Spectra in Magnetic Field

Paschen-Back Effect $\mu_B B \gg \xi$

$$\begin{aligned}\xi_{m_l m_s} &= \frac{\mu_B B_z}{\hbar} \left(\langle \hat{L}_z \rangle_\psi + 2 \langle \hat{S}_z \rangle_\psi \right) \\ &= \frac{\mu_B B_z}{\hbar} (m_l + 2m_s)\end{aligned}$$

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Features

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- L and S precess independently around B
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Landé g-factor

$$g_j = 1 + \frac{j(j+1) + s(s+1) - l(l+1)}{2j(j+1)}$$

Stern-Gerlach Experiment

Boson & Fermions

Hello world

Stern-Gerlach Experiment

Conclusion

Hello world